Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			The {heat / energy} required to {raise / change} [the temperature of] 1 kg (1) by 1 °C or K (1)	2			2		
	(b)	(i)		Mass of the water = $\text{density} \times \text{volume} = 10^3 \times (1.2 \times 10^{-3}) = 1.2\ \text{kg}$ (1) Heat required to increase temperature = shc × mass × increase in temperature = 4200 × 1.2 × (100 – 18) = [413 280 J] (1) Time required = $\frac{\text{heat required}}{\text{power}} = \frac{413\,280}{3 \times 10^3}$ = 137.8 [s] (1) Or 2 min 18 s		3		3	3	
		(ii)		Use of density equation on milk [expect 0.037 kg] (1) Attempt at conservation of energy (1) e.g. Heat lost by the tea = Heat gained by the milk [expect 11 424 and 11 550 [J]] Correct equation set up (1) e.g. $(95-84)(4200)(0.25) = (84-5)(3900)m_{\text{milk}}$ **or** $(95-\theta)(4200)(0.25) = (\theta-5)(3900)m_{\text{milk}}$ **or** equivalent equation set up to calculate one of the variables **or** 0.25 × 4200 × 95 + 0.03708 × 3900 × 5 = (0.25 × 4200 + 0.03708 × 3900) × T (note that this line also works if temperature is in K) Final answer **and** comment (1) e.g. $V = 3.64 \times 10^{-5}\ \text{m}^3$ and so about right **or** final temperature = 84.1 °C and so about right			4	4	3	
				Question 4 total	**2**	**3**	**4**	**9**	**6**	**0**